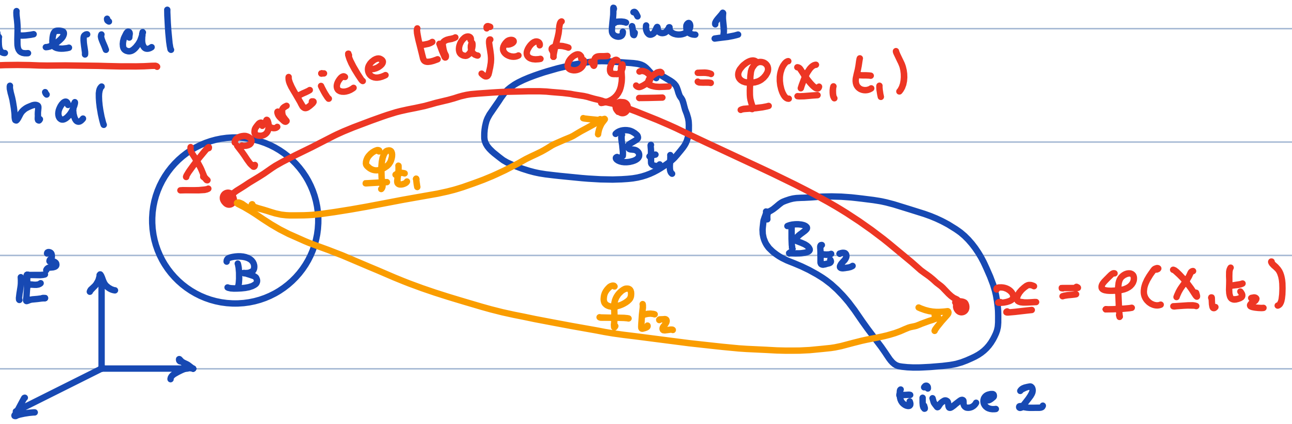
Motion

B: • undeformed
• reference
• material
• initial

B_t: • deformed
• spatial
• current



Continuous deformation of a body over time

$\Rightarrow$ motion: $\quad \underline{x} = \varphi(\underline{X}, t)$

Inverse motion: $\quad \underline{X} = \psi(\underline{x}, t)$

Two types of time derivatives:

1) spatial/local: $\boxed{\left.\frac{\partial \phi}{\partial t}\right|_{\underline{x}} = \frac{\partial \phi}{\partial t}}$ (keep $\underline{x}$ fixed)

rate of change of ϕ as seen by an observer at $\underline{x}$.

2) material/total: $\boxed{\left.\frac{\partial \phi}{\partial t}\right|_{\underline{X}} = \dot{\phi}}$ (keep $\underline{X}$ fixed)

rate of change of ϕ seen by an observer following the path line of a particle.

Velocity and acceleration fields

The velocity and acceleration of a material particle labeled $\underline{X}$ in $\mathcal{B}$ at time t due to motion $\varphi(\underline{X},t)$

$$\boxed{\begin{aligned} \underline{V}(\underline{X},t) &= \dot{\varphi}(\underline{X},t) = \frac{\partial}{\partial t}\varphi(\underline{X},t) = \frac{\partial \underline{x}}{\partial t}\Big|_{\underline{X}} \\ \underline{A}(\underline{X},t) &= \dot{\underline{V}}(\underline{X},t) = \ddot{\varphi}(\underline{X},t) = \frac{\partial^2}{\partial t^2}\varphi(\underline{X},t) = \frac{\partial^2 \underline{x}}{\partial t^2}\Big|_{\underline{X}} \end{aligned}}$$

Easy because $\varphi = \varphi(\underline{X},t) \rightarrow$ material field

Spatial descriptions: $\underline{X} = \underline{\psi}(\underline{x},t)$

$$\left.\begin{aligned} \underline{v}(\underline{x},t) &= \dot{\varphi}(\underline{X},t)\Big|_{\underline{X}=\psi(\underline{x},t)} \\ \underline{a}(\underline{x},t) &= \ddot{\varphi}(\underline{X},t)\Big|_{\underline{X}=\psi(\underline{x},t)} \end{aligned}\right\} \text{substitute after differentiation!}$$

The spatial fields $\underline{v}$ and $\underline{a}$ correspond to the material particle whose current coordinates are $\underline{x}$ at time t.

Note:

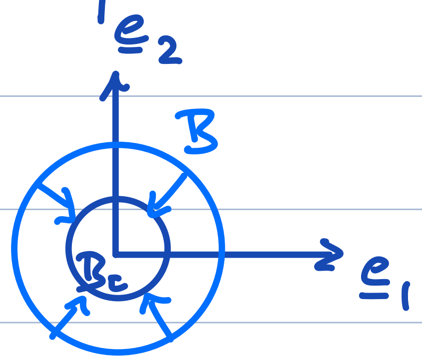
material field: $M(\underline{X},t) \longrightarrow M(\psi(\underline{x},t),t) \equiv M_s(\underline{x},t)$

spatial representation

spatial field: $s(\underline{x},t) \longrightarrow s(\varphi(\underline{X},t),t) \equiv s_m(\underline{X},t)$

material representation

Example: Shrinking shere



$$\varphi(\underline{X},t) = e^{-\lambda t}\,\underline{X} \qquad \lambda > 0$$

$$\psi(\underline{x},t) = e^{\lambda t}\,\underline{x}$$

Velocity & acceleration:

$$\left.\begin{aligned} \underline{V}(\underline{X},t) &= \dot{\underline{\varphi}}(\underline{X},t) = -\lambda e^{-\lambda t}\,\underline{X} \\ \underline{A}(\underline{X},t) &= \dot{\underline{V}}(\underline{X},t) = \lambda^2 e^{-\lambda t}\,\underline{X} \end{aligned}\right\} \text{ material fields}$$

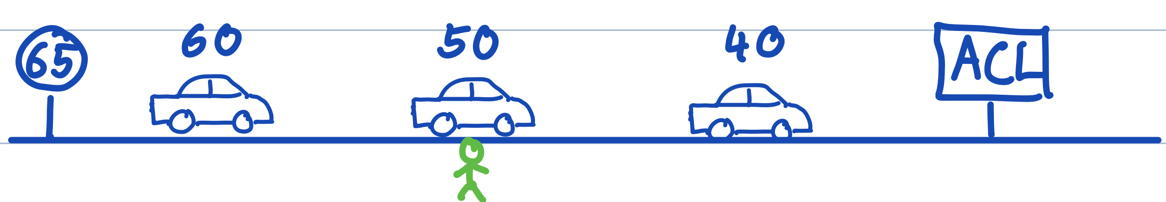
spatial representation $\Rightarrow$ $\underline{X} = \psi(\underline{x},t) = e^{\lambda t}\,\underline{x}$

$$\underline{v}(\underline{x},t) = \underline{V}(\underline{\psi}(\underline{x},t),t) = -\lambda e^{-\lambda t}\left(e^{\lambda t}\,\underline{x}\right) = -\lambda\,\underline{x}$$

$$\underline{a}(\underline{x},t) = \underline{A}(\underline{\psi}(\underline{x},t),t) = \lambda^2 e^{-\lambda t}\left(e^{\lambda t}\,\underline{x}\right) = \lambda^2\,\underline{x}$$

Note: $\frac{\partial \underline{v}}{\partial t} = \underline{0} \neq \underline{a}$ $\Rightarrow$ $\boxed{\underline{a} \neq \frac{\partial \underline{v}}{\partial t}}$

spatial acceleration is not local derivative of spatial velocity field.



each car is decelerating as it approaches ACL: $\underline{a}(\underline{x},t) < 0$

all cars go $\approx$ 50 mph when they pass green person: $\dot{\underline{v}}(\underline{x},t) \approx 0$

Compute material derivative of spatial field?

Example: $T(\underline{x},t) = \alpha t |\underline{x}| e^{\lambda t}$

$$\boxed{\dot{T}(\underline{x},t) = \underbrace{\frac{\partial}{\partial t} T(\underline{\varphi}(\underline{X},t),t)\Big|_{\underline{X}=\underline{\psi}(\underline{x},t)}}_{\mathrm{II}} = \underbrace{\frac{\partial}{\partial t} T_m(\underline{X},t)\Big|_{\underline{X}=\underline{\psi}(\underline{x},t)}}_{\mathrm{I}}}$$

Note: spatial representation of material derivative of a spatial field.

I) Differentiate material representation

- pull it back to material configuration

motion: $\underline{x} = \underline{\varphi}(\underline{X},t) = e^{-\lambda t}\underline{X}$

$T(\underline{\varphi}(\underline{X},t),t) = \alpha t |e^{-\lambda t}\underline{X}| e^{\lambda t} = \alpha t |\underline{X}|$

$T_m(\underline{X},t) = \alpha t |\underline{X}|$ material representation

- material time derivative:

$\dot{T}_m = \frac{\partial}{\partial t} T_m(\underline{X},t) = \alpha |\underline{X}|$

- push forward to spatial configuration

inverse motion: $\underline{X} = \underline{\psi}(\underline{x},t) = e^{\lambda t}\underline{x}$

$\dot{T}_m\Big|_{\underline{X}=\underline{\psi}(\underline{x},t)} = \alpha |e^{\lambda t}\underline{x}| = \alpha e^{\lambda t} |\underline{x}| = \dot{T}(\underline{x},t)$

$\underline{\dot{T}(\underline{x},t) = \alpha e^{\lambda t} |\underline{x}|}$

What if we don't have φ and ψ?

II) Use chain rule

$\Rightarrow$ 2 time dependencies $T(\varphi(\underline{X},t),t)$ & keep $\underline{X}$ fixed!

Quick version:

$$\dot{T}(\underline{x}(t),t) = \frac{\partial T}{\partial t} + \frac{\partial T}{\partial x_i}\underbrace{\frac{\partial x_i}{\partial t}}_{v_i} = \frac{\partial T}{\partial t} + v_i \frac{\partial T}{\partial x_i} = \frac{\partial T}{\partial t} + \underline{v}\cdot\nabla T$$

Careful version (keeping track of $\underline{X}$)

$$\dot{T}(\underline{x},t) = \frac{\partial}{\partial t} T(\underbrace{\varphi(\underline{X},t)}_{\underline{x}},t)\Big|_{\underline{X}=\psi(\underline{x},t)}$$

$$= \underbrace{\left[\frac{\partial T}{\partial t}\Big|_{\underline{x}=\varphi(\underline{X},t)} + \frac{\partial T}{\partial x_i}\Big|_{\underline{x}=\varphi(\underline{X},t)} \frac{\partial}{\partial t}\varphi_i(\underline{X},t)\right]}_{\dot{T}_m(\underline{X},t)\ \text{material representation}}\Bigg|_{\underline{X}=\psi(\underline{x},t)}$$

velocity: $\frac{\partial \varphi_i}{\partial t} = V_i(\underline{X},t) = v_i(\underline{x},t)\big|_{\underline{x}=\varphi(\underline{X},t)}$

$$\dot{T}_m(\underline{X},t) = \frac{\partial T}{\partial t}(\underline{x},t) + \underbrace{\frac{\partial T}{\partial x_i} v_i(\underline{x},t)}_{\underline{v}(\underline{x},t)\cdot\nabla T(\underline{x},t)}\Big|_{\underline{x}=\varphi(\underline{X},t)}$$

$$\dot{T}(\underline{x},t) = \frac{\partial T}{\partial t} + \underline{v}\cdot\nabla T$$

Example: $T(\underline{x},t) = \alpha t e^{\lambda t} |\underline{x}|$

$$\underline{v}(\underline{x},t) = -\lambda \underline{x}$$

$$\frac{\partial T}{\partial t} = \alpha |\underline{x}| \left(e^{\lambda t} + \lambda t e^{\lambda t} \right) = \alpha |\underline{x}| (1+\lambda t) e^{\lambda t}$$

$$\nabla T = T_{,i} \underline{e}_i = \alpha t e^{\lambda t} (x_k x_k)^{1/2}{}_{,i} \underline{e}_i$$

$$= \alpha t e^{\lambda t} \frac{1}{2} (x_k x_k)^{-1/2} 2 (x_{k,i} x_k) \underline{e}_i$$

$$= \alpha t e^{\lambda t} \frac{1}{|\underline{x}|} \delta_{ki} x_k \underline{e}_i = \alpha t e^{\lambda t} \frac{\underline{x}}{|\underline{x}|}$$

$$\dot{T} = \frac{\partial T}{\partial t} + \underline{v} \cdot \nabla T$$

$$= \alpha |\underline{x}| (1+\lambda t) e^{\lambda t} - \lambda \underline{x} \cdot \left(\alpha t e^{\lambda t} \frac{\underline{x}}{|\underline{x}|} \right)$$

$$- \alpha \lambda t e^{\lambda t} \frac{\underline{x} \cdot \underline{x}}{|\underline{x}|}$$

$$= \alpha |\underline{x}| e^{\lambda t} (1 + \lambda t - \lambda t)$$

$\dot{T} = \alpha |\underline{x}| e^{\lambda t}$ same as method I !

Spatial rep. of mat. deriv. of spatial field:

$$\boxed{\dot{\phi} = \frac{\partial \phi}{\partial t} + \nabla \phi \cdot \underline{v}} \quad \text{and} \quad \boxed{\dot{\underline{w}} = \frac{\partial \underline{w}}{\partial t} + (\nabla \underline{w}) \underline{v}}$$

The result for $\dot{\underline{w}}$ follows by applying the scalar result to w_i.

This result is important because it allows the computation of $\dot{\phi}$ and $\dot{\underline{\omega}}$ without knowledge of φ, if the velocity is known!

$\Rightarrow$ in fluid mechanics you never see φ

In fluid mechanics: $(\nabla \underline{\omega})\underline{v} = \underline{v} \cdot \nabla \underline{\omega}$ see HW7!

The spatial acceleration field is defined as

$$\boxed{\underline{a} = \dot{\underline{v}} = \frac{\partial \underline{v}}{\partial t} + (\nabla^{x} \underline{v})\,\underline{v}}$$

Spatial acceleration is a non-linear function of spatial velocity $\Rightarrow$ basic non-linearity in fluid mechanics

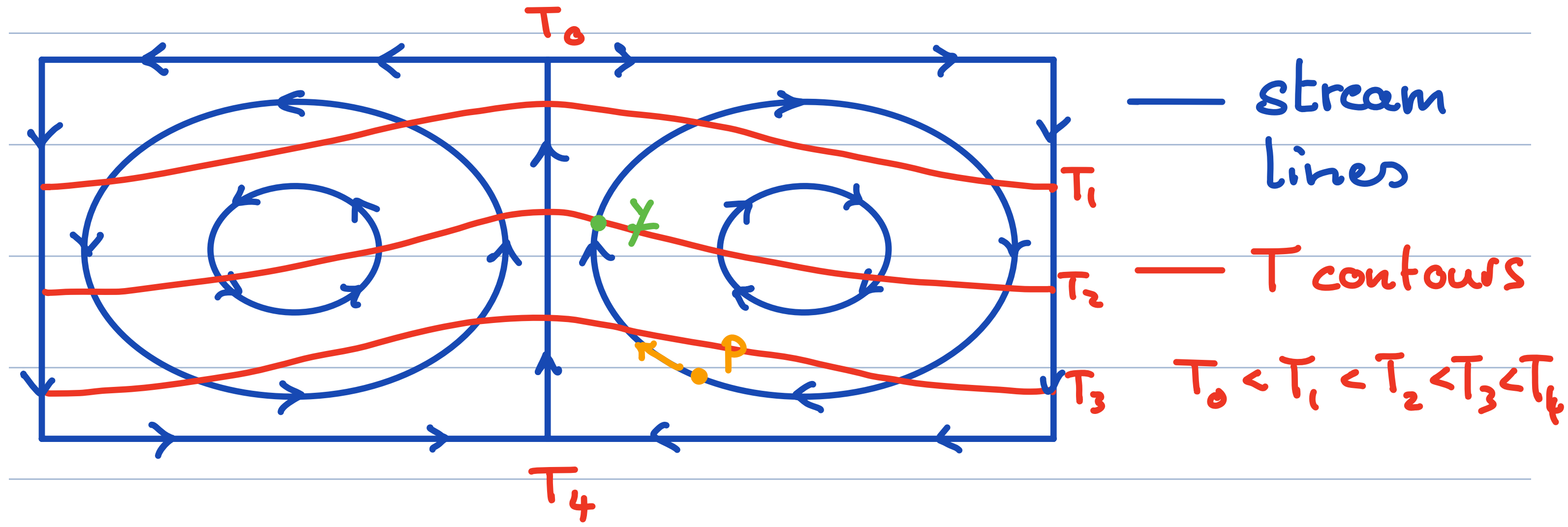
Example: $\underline{v} = -\lambda \underline{x} \qquad \underline{a} = \lambda^2 \underline{x}$

$$\underline{a} = \cancel{\frac{\partial \underline{v}}{\partial t}}^{\,0} + (\nabla \underline{v})\,\underline{v} \qquad \nabla \underline{v} = v_{i,j}\, \underline{e}_i \otimes \underline{e}_j$$

$$\underline{a} = -\lambda \underline{\underline{I}}(-\lambda \underline{x}) \qquad v_{i,j} = -\lambda x_{i,j} = -\lambda \delta_{ij}$$

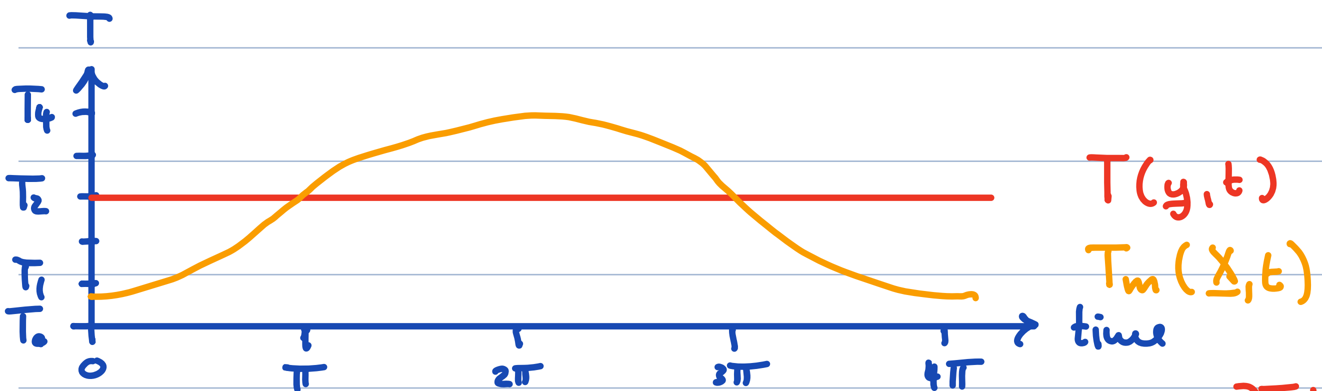
$$\underline{a} = \lambda^2 \underline{x} \checkmark \qquad \Rightarrow \nabla \underline{v} = -\lambda \underline{\underline{I}}$$

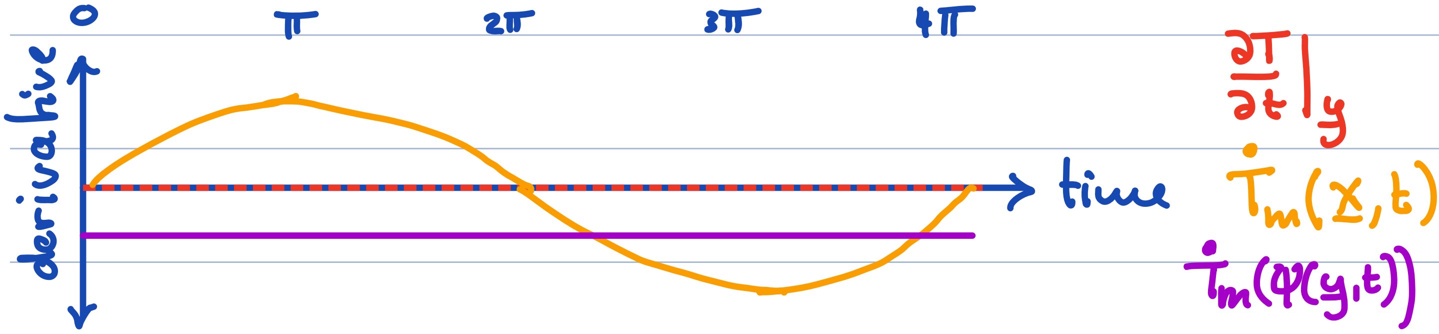
Example: Steady convection



Steady state: $T(\underline{x}, t) = T(\underline{x})$ (spatial field)

Clearly, $T(\underline{y}) = T_2$ and $\left.\frac{\partial T}{\partial t}\right|_{\underline{y}} = 0$





Consider a particle P with initial location $\underline{X}$

and current position $\underline{x} = \varphi(\underline{X}, t)$, which passes through y once every overturn.

Its temperature is $T(\varphi(\underline{X},t)) = T_m(\underline{X},t)$ is a material field and oscillates periodically. Its derivative the "material derivative" is $\dot{T}_m(\underline{X},t) \neq 0$